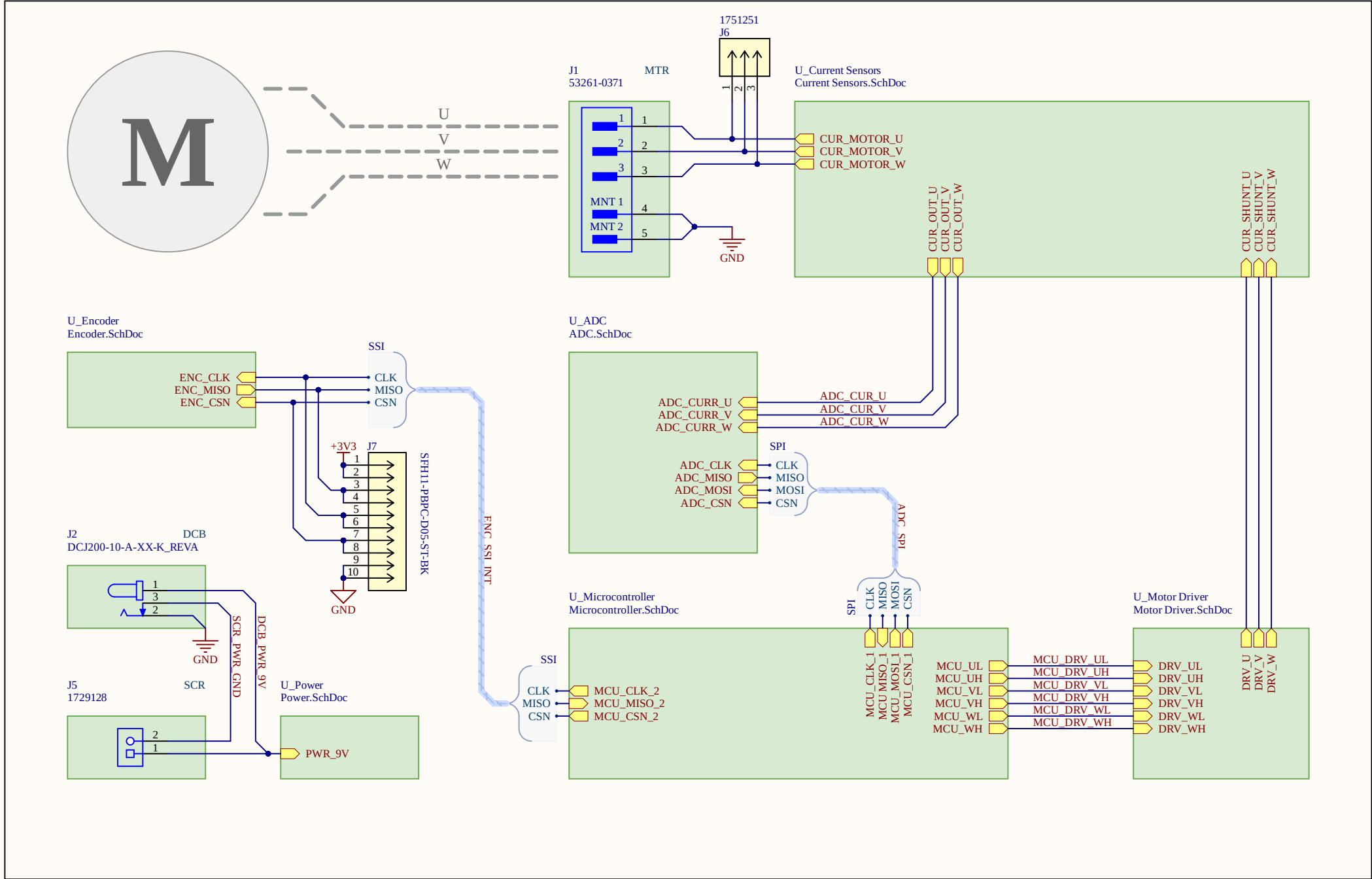


HAPTIC KNOB - FOR SIMULATING ELECTRICAL PHENOMENA

Description.

UBC Open Robotics
2025/2026



Legend:
MCU = Microcontroller
PWR = Power System
DRV = Motor Driver
CUR = Current Sensor
ADC = Analog to Digital Converter
ENC = Encoder
DCB = DC Barrel Jack
MTR = Motor Connector
SCR = Screw-In Terminal

Patrick Delfin, Harry Nguyen, Trieu Truong

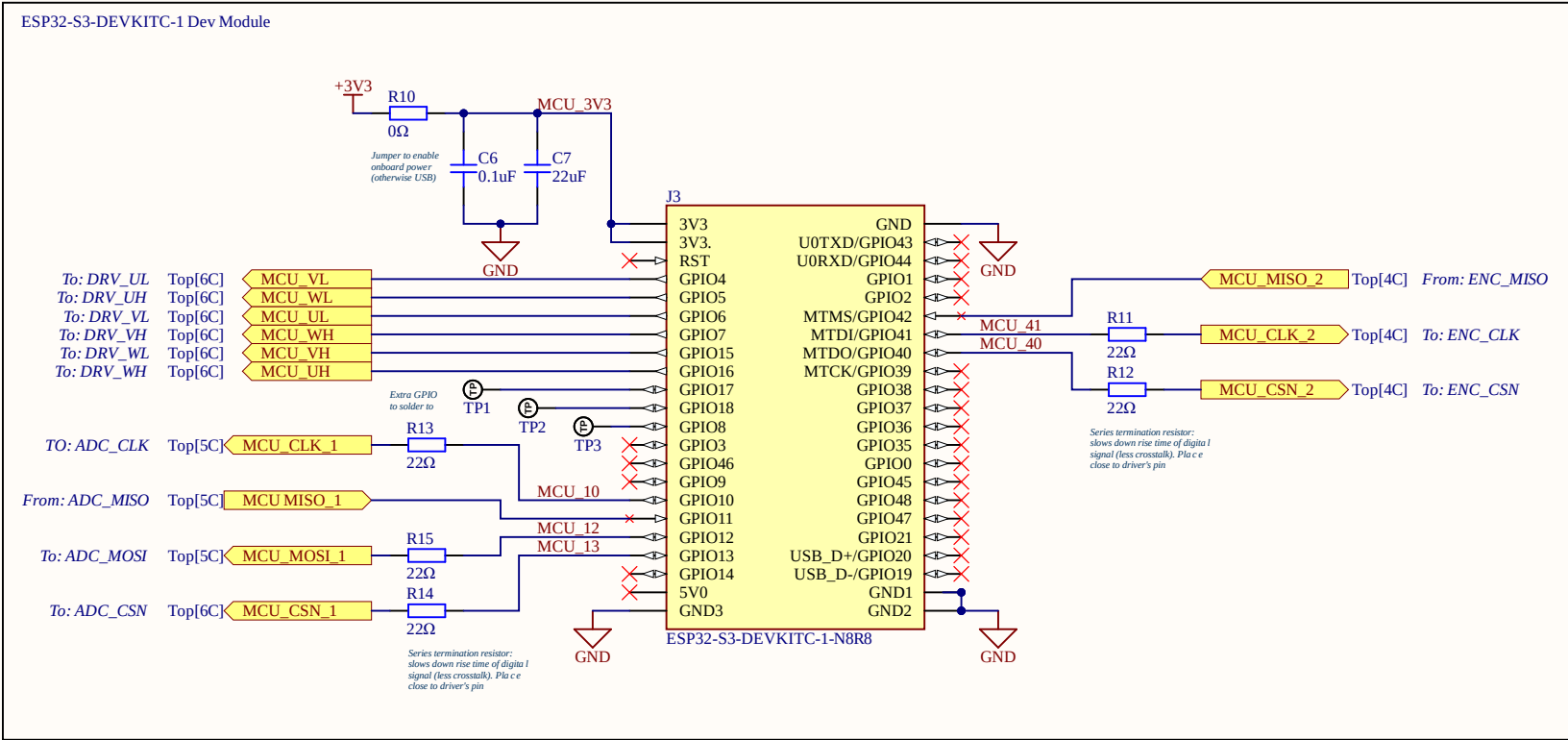


Title		
Haptic Knob Subteam - UBC Open Robotics		
Size	Number	Revision
A3		
Date:	3-18-2026	Sheet of
File:	C:\Users\...\Top.SchDoc	Drawn By:

MICROCONTROLLER

Executes the control system including reading sensors, controlling the motor driver, and performing the haptics calculations.

Harry Nguyen
2026-02-12



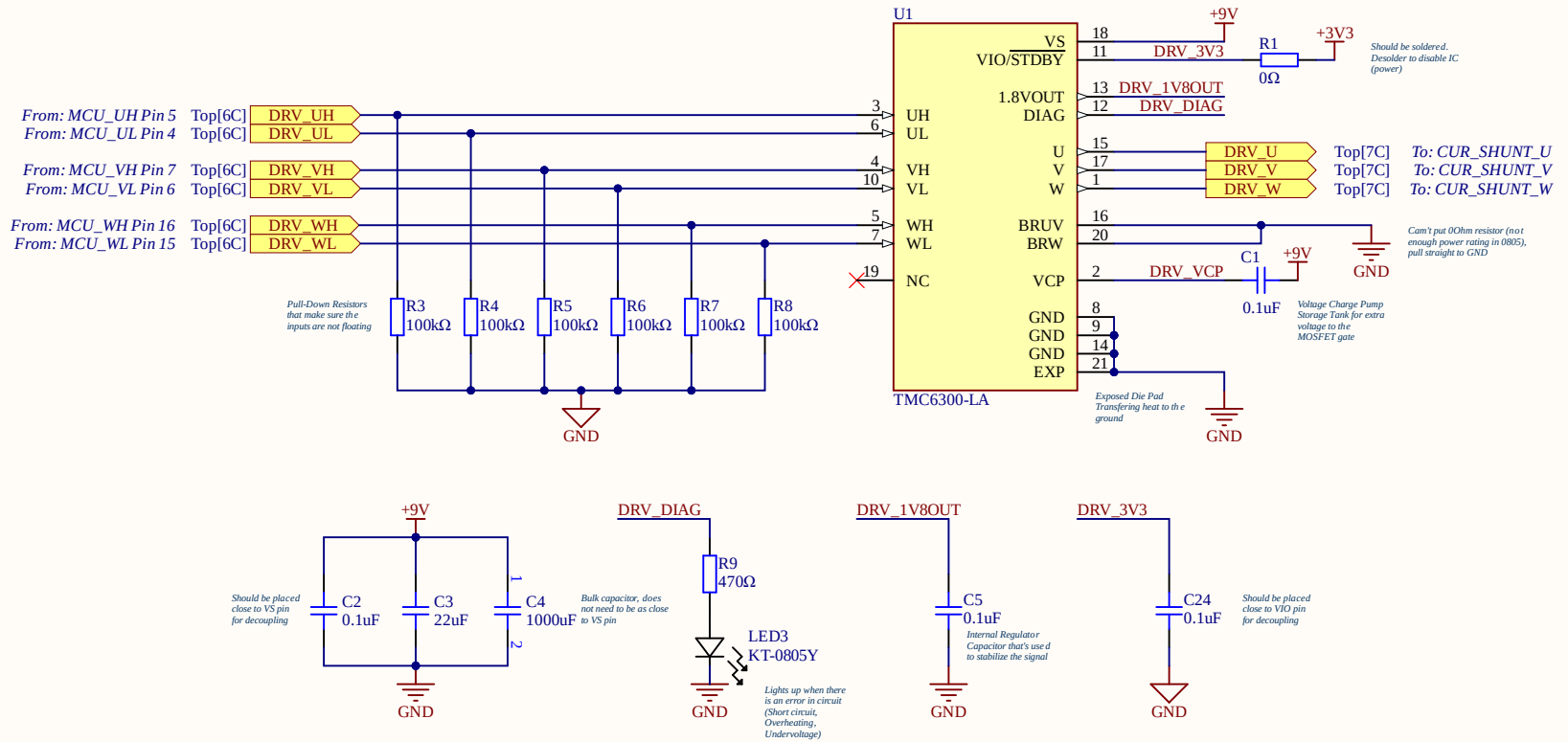
Title Haptic Knob Subteam - UBC Open Robotics		
Size A4	Number	Revision
Date: 3-18-2026	Sheet of	
File: C:\Users\...\Microcontroller.SchDoc	Drawn By:	

MOTOR DRIVER

Provides the necessary high current and voltage to manage speed, direction, and torque.

Trieu Truong
2026-01-31

TMC6300 3-Phase BLDC Motor Driver



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SENSOR - CURRENT SENSORS

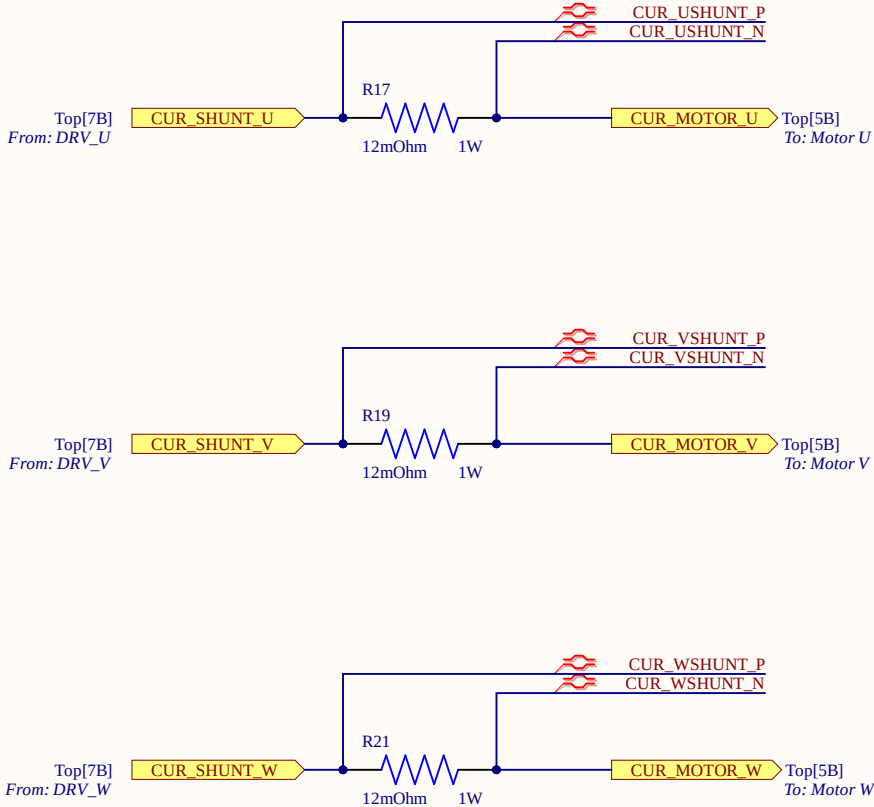
Measures the current through each of the 3 phases by measuring the voltage drop across shunt resistors.

Harry Nguyen
2026-02-28

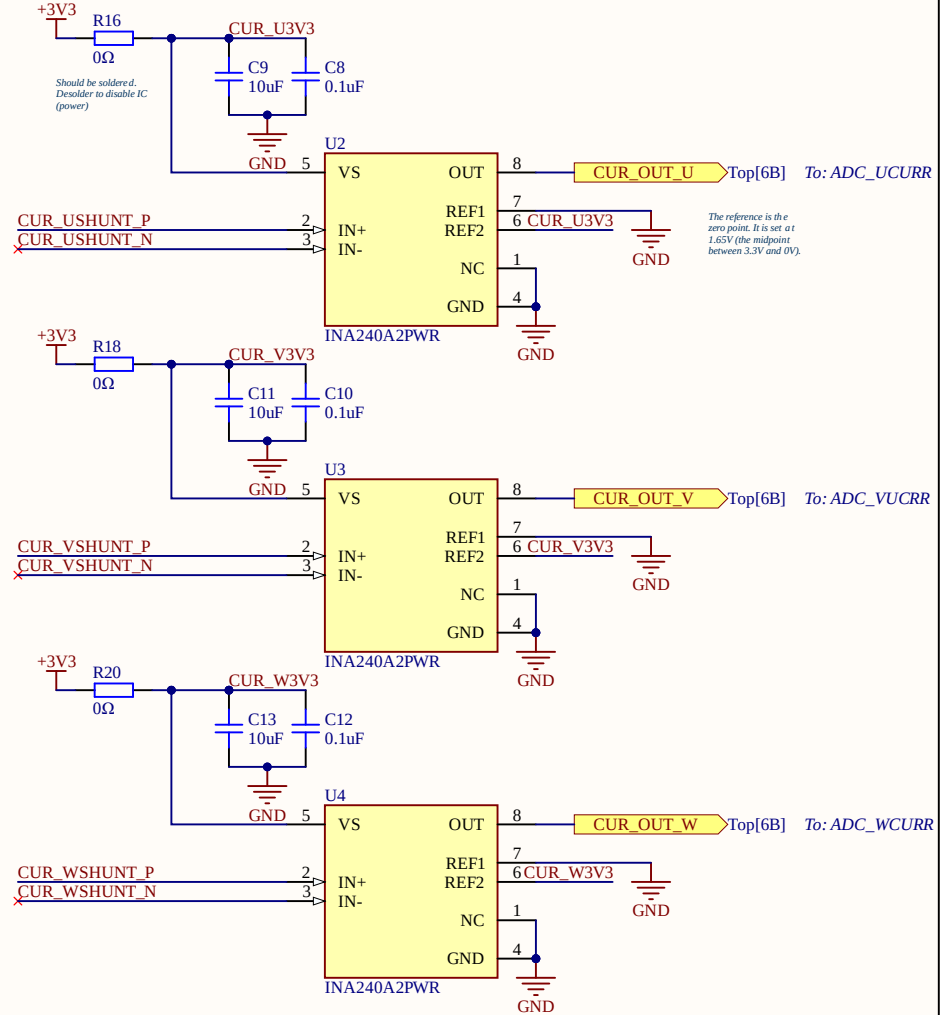
Shunt Resistors (U, V, W)

Symbols denote Differential Pairs, to enable CMNR between the signal pair U_SHUNT_P and U_SHUNT_N

- Traces consideration:
+ Parallel for the entire length
+ Must be the same length
+ Must be as short as possible (long traces can act like antennas that pick up noise)
+ Route the wires as close together as possible: routing the lines close together, any external magnetic field will induce the same noise voltage in both wires. The current sense amplifier looks at the difference between the two wires, so the noise (which is identical on both) gets cancelled out mathematically.



INA240A2 Current Sense Amplifiers (U, V, W)



Notes:
INA240A2 Gain: 50V/V

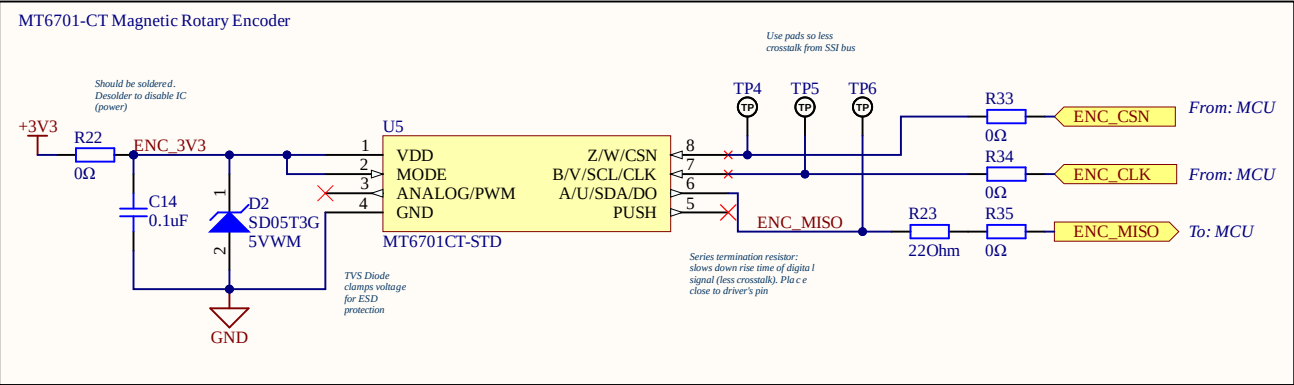
Title Haptic Knob Subteam - UBC Open Robotics		
Size A4	Number	Revision
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File: C:\Users\...\Current Sensors.SchDoc	Drawn By:	



SENSOR - MAGNETIC ROTARY ENCODER

Reads the absolute angle of the motor's rotor.

Patrick Delfin
2026-01-23



Notes:
Configured Output: SSI (SPI)
Powered By: 3.3V linear voltage regulator
Max RPM: 55,000 RPM
Package: SOP-8

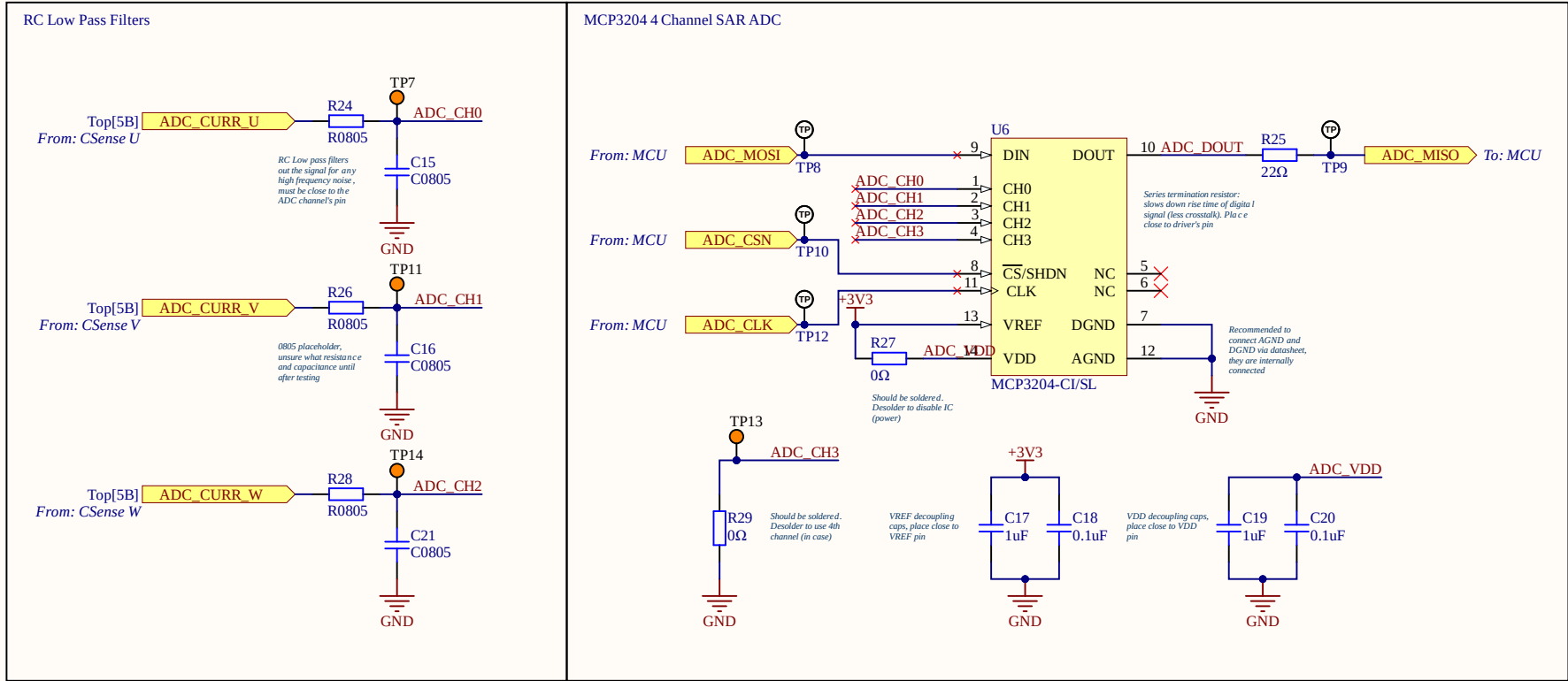


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Size A4	Number	Revision
Date:	3-18-2026	Sheet of
File:	C:\Users\...\Encoder.SchDoc	Drawn By:

ANALOG TO DIGITAL CONVERTER

Converts analog current reading to digital SPI protocol.

Patrick Delfin
2026-02-04



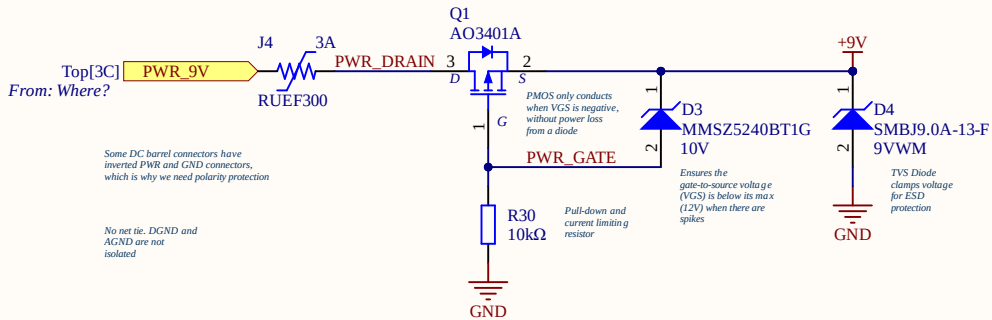
Title Haptic Knob Subteam - UBC Open Robotics		
Size A4	Number	Revision
Date: 3-18-2026	Sheet of	
File: C:\Users\...\ADC.SchDoc	Drawn By:	

POWER SYSTEM

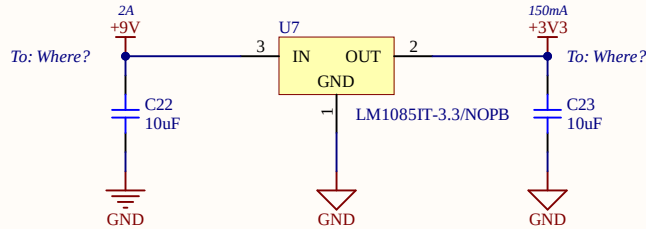
Supplies a protected 9V and 3.3V to the entire system.

Patrick Delfin
2026-02-12

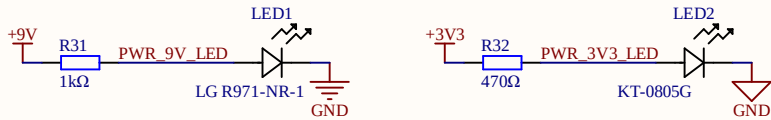
Fuse, Reverse Polarity Protection, and TVS Diode



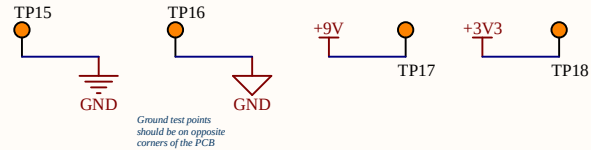
LM1085 3.3V Fixed LDO Voltage Regulator



Power Indicator LEDs



Power Test Points



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Haptic Knob Subteam - UBC Open Robotics

Size	Number	Revision
A4		

Date:	3-18-2026	Sheet of
File:	C:\Users\...\Power.SchDoc	Drawn By: